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free-radical Bulk Polymerization Inhibitors**

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Abstract

Noni (*Morinda citrifolia*) roots are a good source of anthraquinones, such as damnacanthal, nordamnacanthal and morindone, which have demonstrated *in vitro* cytotoxic and antitumor activity. In this work, the inhibitory properties of anthraquinones contained in Noni root extracts against styrene free-radical bulk polymerization were studied. The extraction methodology was based on Singh *et al.* method where isolation of some anthraquinones was achieved. Styrene polymerizations were performed at 60 °C with the presence of AIBN as initiator (blank) at 1% weight and with two extracts of Noni roots at 1% weight. Inhibitory capacity, expressed as conversion of styrene, was compared with hydroquinone. Chloroformic root extract exhibited the highest inhibitory capacity with an inhibition time between 15 min to 30 min, thus exceeding hydroquinone in which the inhibition time was less than 15 min. The ethanolic extract was not as efficient because it contained lower concentrations of anthraquinones. Damnacanthal was isolated and characterised from chloroformic root extract with H-NMR spectroscopy.

1. Introduction

Noni (*Morinda citrifolia* L.) is a plant native to tropical Asia and Oceania, member of the Rubiaceae family and is extensively distributed in the Pacific region. The plant is an evergreen small tree that grows to about 3-10 m tall [1]. Roots are the most important source of anthraquinones in the plant [2].

Anthraquinones have gained interest as phytochemicals with anti-tumor and anti-cancer properties. Some of these anthraquinones such as damnacanthal, nordamnacanthal and morindone have shown inhibitory effects of cancer cells *in vitro* [3-6]. It has also been determined that these compounds possess antioxidant and radical scavenging properties [7-10].

In this work we evaluated the use of anthraquinones as inhibitors of styrene free radical bulk polymerization in order to find more efficient and cheaper inhibitors than the used commercially.

2. Methodology

2.1 *Plant material*

Noni roots were harvested in the department of Carazo, Nicaragua, from two Noni plants five years old, cultivated at 400 m above sea level. Roots were washed with tap water in order to remove soil remains and then chopped into small pieces. A disc mill was used to reduce root size until a sawdust-like material was obtained.

Roots were then sun dried the next two days for 8 hours each day. Root powder was then packed and sent to the Research Center in Applied Chemistry (CIQA) in Saltillo, Mexico for experimentation.

2.2 *Extraction procedure*

A sample containing 100 g of dried root powder was extracted with ethanol in a laboratory set up consisting in a stirring hot plate, a round flask immerse in an oil bath with a condenser on top. 500 mL of absolute ethanol (Merck Chemicals) was added to the system which kept continuously operating for 16 hours at 80 °C.

Ethanol phase was then filtered and the root sample was successive extracted in two more batches with 250 mL of ethanol each for the same conditions and time as in the first step. Ethanol collected from the three extraction steps was placed in a rotatory evaporator with vacuum in order to remove the solvent from the extract.

A solid residue of 5.68 g of root extract (ET) was obtained after vacuum distillation of ethanol and was stored in a freezer.

A sample of 4.09 g of this extract was macerated with 100 mL of reagent grade hexane. The mixture was stirred two hours. After this time it was filtered in order to separate the hexane soluble fraction from the insoluble solid. The n-hexane soluble fraction was discarded.

A 1:1 volume mixture of distilled water and analysis grade chloroform (Merck Chemicals) was prepared with 100 mL of each solvent. The hexane insoluble solid obtained from last step was dissolved in this mixture and then poured in a separatory funnel forming two insoluble phases. The mixture was allowed to stand for two hours until the two phases separated completely.

The denser chloroform phase (70 mL) was decanted and separated from the aqueous phase and was dried in a rotary evaporator to obtain 560 mg of a solid residue (CHL). This extraction and separation methodology was based on Singh *et al.*, where the isolation of damnacanthol was achieved [11]. The major antifungal activity was reported in the chloroform extract with this procedure.

2.3 *Description of experiments*

Polymerizations with styrene were carried out at times from 15 min to 15 hours at 60° C in order to measure the styrene conversion. Initially, polymerizations with 99% weight of styrene

and 1% of AIBN (azobisisobutyronitrile) as initiator were carried out as a reference, i.e., without the presence of inhibitors or Noni extract.

Then, polymerizations with 98% of styrene, 1% of AIBN and 1% of Noni root extract were carried out, considering the ethanolic (ET) and chloroform (CHL) solid-soluble extracts. Other set of polymerizations considering 98% of styrene, 1% of AIBN and 1% of Hydroquinone (HQ) were also developed.

2.4 Isolation of anthraquinones

Chloroformic root extract (CHL) was chromatographed on silica gel column, eluted with a mixture of 60% chloroform, 38% petroleum ether and 2% ethanol. A sample of this extract was analyzed in Gel Permeation Chromatography (GPC) to determine the molecular weights of compounds.

Twenty seven different fractions were collected in several vials corresponding to different color interphases in the chromatographic elution. Each fraction was vacuum dried and the solid crystals were stored in a freezer at 4 °C until analysis. H NMR spectra was obtained for some fractions, in order to detect anthraquinone presence in the samples.

Some of the collected fractions were analyzed in a Fourier Transform spectrophotometer (FTIR) to identify the functional groups and structure of the compounds. Fractions that exhibited a defined spectra were purified by recrystallization and reanalyzed in proton NMR and FTIR.

3. Results and discussion

3.1 Effect of extracts on styrene conversion

The results of polystyrene formation for each set of experiments are shown in the following table:

Table 1. Comparison of conversion vs. time for different polymerizations

Time (min)	Styrene Conversion (%)			
	No Inhibitor	Hydroquinone	Chloroformic Extract	Ethanolic Extract
15	2.7	2.1	0.0	2.0
30	3.4	2.9	2.4	3.1
60	6.6	4.9	3.1	7.0
120	12.7	11.7	11.3	13.4
300	29.6	18.3	25.8	25.9
600	62.7	38.2	51.1	55.0
900	91.3	67.6	87.7	82.6

The chloroformic root extract prevented the formation of polystyrene until 15 min, exhibiting an inhibition time between 15 to 30 min. Hydroquinone and ethanolic extract had no inhibition

time or it was less than 15 min, according to reaction conditions. The styrene conversion induced by chloroformic extract was the lowest for short times (0 – 120 min).

Ethanollic extract exhibited a polystyrene formation close to reference. Hydroquinone reduced the conversion of styrene but its performance was lower than chloroformic extract.

At higher times (300 – 900 min) hydroquinone showed an inhibitor/retardant behavior, reducing polystyrene formation more than ethanollic and chloroformic root extract.

3.2 Effect of extracts on polystyrene molecular weight distribution

The molecular weight distribution curves of polystyrene produced in the experiments are shown in Fig. 1.

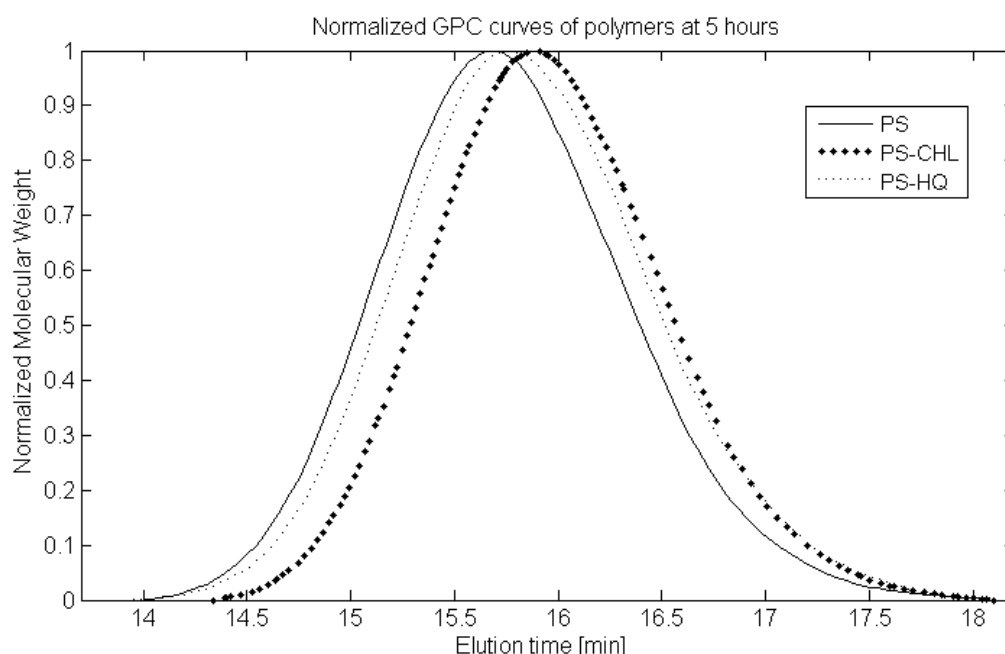


Figure 1: GPC of polystyrene obtained at 5 hours. PS: 99% styrene, 1% AIBN; PS-CHL: 98% styrene, 1% AIBN, 1% chloroformic root extract; PS-HQ: 98% styrene, 1% AIBN, 1% hydroquinone.

The molecular weight distribution of polystyrene in presence of chloroformic root extract (PS-CHL) was displaced to the right (equivalent to lower molecular weights) compared with both hydroquinone (PS-HQ) and reference (PS).

Hydroquinone caused a lesser displacement to lower molecular weights and the polydispersity index of the PS-HQ was higher (1.819) than PS-CHL (1.618), meaning a broader molecular weight distribution of polystyrene for PS-HQ.

At times of 10 and 15 hours the same trend remained, chloroformic root extract was more effective in reducing the average molecular weight than hydroquinone and its polydispersity index was the lowest.

3.3 Characterization of anthraquinones from *Noni* extracts

¹H NMR characterization in different fractions obtained from silica gel column chromatography revealed the predominant presence of anthraquinones, especially in chloroformic *Noni* root extract.

Signals from Fig. 2 were compared with those reported by Faltynek *et al.* [12] for damnacanthal, with a positive identification.

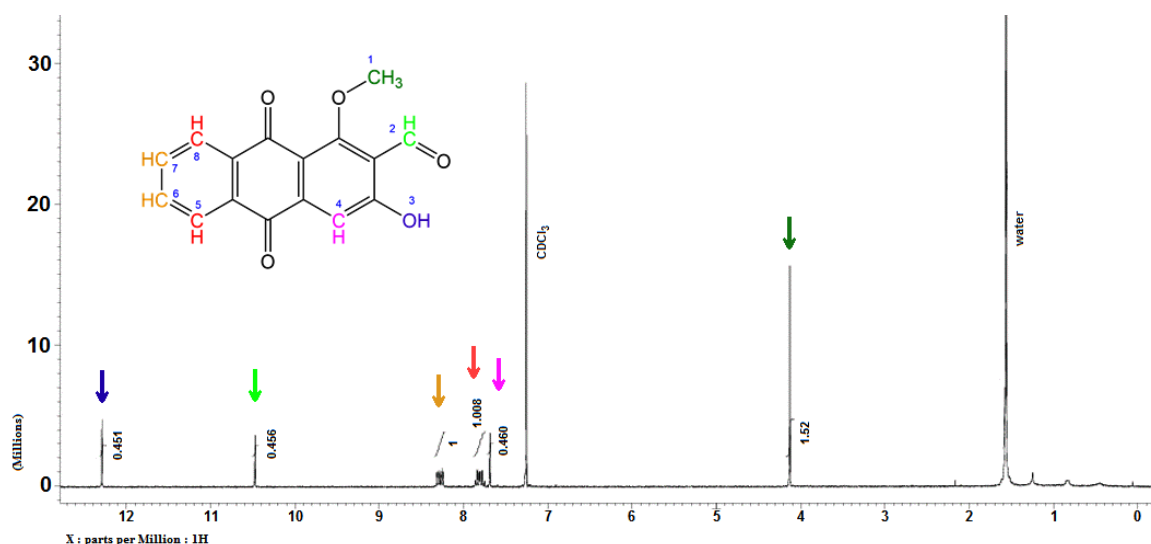


Figure 2: ¹H NMR Spectrum (400 MHz) in CDCl₃ obtained of Damnacanthal (anthraquinone) isolated from *Noni* chloroformic root extract (CHL) by silica gel column chromatography.

NMR and FTIR spectra confirmed the typical structure of fused benzene rings and ketone groups at carbons 9 and 10 from anthraquinones and some functional groups like: OCH₃, CHO and OH; characteristic of anthraquinones with important biological activity as damnacanthal, alizarin, nordamnacanthal, and scopolamine.

The GPC of chloroformic root extract indicated a molecular weight distribution between 268 and 353 g/mol, with an average molar molecular weight of 289 g/mol, placing these compounds in the molecular weight range of anthraquinones.

4. Conclusions

The chloroformic root extract (CHL) performed better as an inhibitor than hydroquinone (HQ) and ethanolic extract (ET), inducing the lowest styrene conversion at short times (0 – 120 min) with an inhibition time between 15 to 30 min. Hydroquinone and ethanolic extract exhibited no inhibition time at reaction conditions.

Chloroformic root extract also induced the lowest polydispersity index in polystyrene formation, meaning a smaller range of molecular weight and a better performance. It also generated a lower molecular weight distribution of polystyrene compared to hydroquinone and ethanolic extract.

Damnacanthal, an anthraquinone with anticancer properties [4-6,12], was isolated from chloroformic root extract. FTIR, GPC and ¹H NMR spectra of fractions separated by silica gel column chromatography confirmed the majoritarian presence of anthraquinones in this extract.

Considering that hydroquinone and AIBN were added pure to the bulk polymerization of styrene and that only some of the compounds from the Noni root extracts had radical scavenging properties, the performance of chloroform root extract as inhibitor was remarkable. This means that anthraquinones from Noni roots might be a good source of inhibitors against radical polymerization with a better performance than common inhibitors like hydroquinone.

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